Review notes · for you

Patterns in data: unit check

Data science · Level 2 · Unit check

Time	Materials
25 min	Paper and a pen. No device needed.

Learning goals

- Histograms and shapes
- Spread and outliers
- Scatter and correlation
- A routine for exploring

Activity: Revise in pairs

1. Pairs (2 min): Work with a partner and share one sheet for your answers.
2. Quiz (15 min): Read each question, agree on an answer and say why.
3. Check (8 min): Compare with the answers and redo the ones you got wrong.

Discussion

- Which idea will you use first?
- Which table will you explore?
- Which claim will you question?

Check yourself

- One finding I will look for this week, and where.

Try it this week

Explore one small table from your own life with the five questions, and share one finding with a number.

Quiz sheet

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Name _____ Class _____ Date _____

1 **What does each bar of a histogram show?**

◯ How many values are in its bin ◯ The mean ◯ One person's value

2 **In right-skewed data the median is usually above the mean.**

◯ True ◯ False

3 **Match each term to its meaning.**

Standard deviation	• • • •	Spread of the middle half
IQR	• • • •	A value far from the rest
Outlier	• • • •	Typical distance from the mean

4 **Use the IQR rule.**

Q1 = 20 and Q3 = 28, so the upper fence is _____ .

Word bank: 36 · 40

5 **You find a sleep time of 25 hours. What is it?**

◯ An impossible value to check and fix ◯ A good finding ◯ The median

6 **Which describe r ≈ −0.8? Choose all that apply.**

☐ Fairly strong ☐ Dots fall from left to right ☐ One causes the other

7 **Correlation proves causation.**

◯ True ◯ False

8 **Put the exploring questions in order.**

☐ Shape

☐ Relationships

☐ What is in it?

☐ Centre and spread

☐ Gaps and oddities

9 **Which is the best finding?**

◯ I use my phone a lot. ◯ My median screen time is 3 hours on weekdays and 5 at weekends.

◯ Phones are bad.

10 **Very wide bins can hide the shape of the data.**

◯ True ◯ False

Check your answers

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1. **How many values are in its bin**

   Bars count values in equal bins.

2. **False**

   The tail pulls the mean above the median.

3. **Standard deviation — Typical distance from the mean**
   IQR — Spread of the middle half
   Outlier — A value far from the rest

   Ways to describe spread and surprises.

4. **Q1 = 20 and Q3 = 28, so the upper fence is 40.**

   IQR = 8, and 28 + 1.5 × 8 = 40.

5. **An impossible value to check and fix**

   A day only has 24 hours.

6. **Dots fall from left to right; Fairly strong**

   Negative and strong, but nothing about cause.

7. **False**

   Something else may cause both.

8. **Top to bottom: 2, 5, 1, 3, 4**

   From the whole table to comparisons.

9. **My median screen time is 3 hours on weekdays and 5 at weekends.**

   Specific, with numbers.

10. **True**

    A few fat bars lump everything together.